

SHUBHAM ANIL KHADE

Generative AI Engineer - Audio, Vision & LLM Systems

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PROFESSIONAL SUMMARY

Generative AI Engineer specializing in Audio AI, Computer Vision and LLM systems, with experience shipping production-grade AI pipelines across speech recognition, speaker diarization, voice cloning and live IoT telemetry. First-author ML researcher (GRENZE, 2025) with a track record of taking systems from prototype to production. Strong in Python, PyTorch, Hugging Face, AWS and full-stack AI delivery.

CAREER HIGHLIGHTS

9+ AI projects shipped end-to-end | 1 first-author research publication | 90%+ CV model accuracy | 5+ live telemetry streams at sub-200 ms latency

TECHNICAL SKILLS

Programming Languages: Python (primary), JavaScript, C, C++, SQL, PHP, HTML, CSS

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, CNNs, Computer Vision, YOLO, Object Detection

Audio & Video AI: Whisper (large-v3), Pyannote 3.1, F5-TTS, XTTS-v2, ElevenLabs, Lip-sync, PySceneDetect

Generative AI & LLMs: Hugging Face, Replicate, Together AI, Ollama, Local LLM Inference, Fine-Tuning, RAG, Prompt Engineering

LLMs Used: GPT, Claude, Gemini, Llama, Qwen, DeepSeek, Gemma

Databases & Cloud: MySQL, Firebase, AWS EC2, AWS IoT Core

IoT & Embedded: Arduino, ESP32, MQTT, RS-485, Firmware, Sensor Fusion

Tools & DevOps: Git, GitHub, Docker, Linux, Postman, Jupyter, VS Code, Selenium, Automation

EXPERIENCE

Generative AI Engineer

Nov 2025 - Present

Framex Studio, Pune, India | Core engineering team (1 of 5)

- Architected a 5-stage movie-dubbing pipeline spanning 5+ Indian languages on feature-length (2+ hr) audio - transcription, diarization, translation, voice-cloned TTS and lip-sync - for studio-grade workflows.
- Implemented speaker diarization with Pyannote 3.1 and ASR with Whisper large-v3, enabling accurate multi-speaker segmentation across long-form audio.
- Engineered timbre-preserving voice cloning (F5-TTS, XTTS-v2) with custom noise-removal, scene detection and spectral enhancement for clean, long-form output.

Graduate Engineer Apprentice - Machine Learning & IoT (Internship)

Feb 2025 - Oct 2025

KatMine Technology Pvt. Ltd., India

- Built a live data pipeline ingesting multi-sensor data over the RS-485 protocol, transforming hardware streams for cloud ingestion and dashboard monitoring.
- Maintained 99%+ system uptime via automated monitoring and debugging across the firmware-backend-cloud stack; collaborated cross-functionally to ship the full IoT solution.

PROJECTS

EVTWIN - Real-Time EV Digital Twin & BMS Intelligence System

2024 - 2025

Final-Year Capstone | Live: temperature-demo.vercel.app | Code: github.com/shubhamkhade1555-creator/Temperature_demo

- Designed an Arduino + ESP32 telemetry node capturing 5+ live data streams (battery voltage, current, temperature, motor RPM, GPS) at a 1-10 Hz sample rate.
- Developed a BMS intelligence layer with MQTT-based cloud telemetry at sub-200 ms latency, enabling cell-level anomaly detection and fleet-wide health analytics.
- Created a cloud dashboard for charging-pattern tracking, driver-behaviour scoring and predictive maintenance, closing the 'no monitoring / no logging / no analytics' gap in Indian EVs.

Stack: Python, Pandas, Arduino, ESP32, MQTT, AWS IoT Core, Time-Series Analysis, Anomaly Detection.

- Shipped a browser-based tool for AI-powered audio synchronization, aligning and processing audio tracks entirely client-side.
- Coded the audio-alignment logic in JavaScript and the Web Audio API; open-sourced and deployed on Vercel.

- Trained a CNN-based real-time gesture-recognition pipeline (ResNet backbone) translating sign language to text from video frames, reaching 90%+ test accuracy at 10-15 FPS on CPU.
- Deployed with OpenCV capture and TensorFlow inference; added a novel haptic vibration-feedback channel for two-way deaf-to-hearing communication.

PUBLICATIONS

- Khade, S. (First Author). "Intelligent Machine Learning Communication System for the Deaf." GRENZE Journal of Engineering & Technology (GIJET), Vol 11, Issue 2, 2025 (Paper ID 6348). Available at [thegrenze.com](#).

CERTIFICATIONS

- AWS Academy Graduate - Getting Started with AWS IoT, AWS Training & Certification, 2025.
- Introduction to Amazon EC2, AWS Training & Certification, 2026.

ACHIEVEMENTS

- Presented "IoT-Enabled Two-Wheeler Accident Detection System" at the State-Level IoT Project Competition (2024-25).

EDUCATION

B.Tech, Computer Science & Engineering (IoT & Cyber Security with Blockchain)

2021 - 2025

Annasaheb Dange College of Engineering & Technology, Ashta | CGPA: 7.1/10

Class XII (HSC), Science - 83% (2021), Dr. Bapuji Salunkhe College, Sangli | Class X (SSC) - 73.6% (2019), Vidyamandir Prashala, Miraj

LANGUAGES

English (Professional) | Hindi (Native) | Marathi (Native)